NLP FINAL PROJECT

Search

LearnLoop

Multimodal RAG study assistant

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Agenda

- The problem & Solution
- Model &Technologies
- LearnLoop Pipeline
- LearnLoop Demo

The Problem

Students struggle with academic content spread across multiple formats:

- Long PDFs and lecture notes
- Recorded lectures
- Large textbooks

Our Solution

Upload any study material:

- PDF
- Image
- Audio
- Video

AI Generates:

- Accurate Answers for your question
- Structured Summaries
- Quiz Questions
- Flashcards

Model & Technologies used

1

Huggingface Model sourcing:

- BAAI/bge-m3
- Qwen/Qwen2.5-7B-Instruct

2

Building the MVP:

- PDF upload
- Summary & Quiz generation
- Question Answering

- Whisper base
- DocumentOCR
- FAISS

3

Adding advanced features:

- Flashcards
- Video transcription
- OCR based pdf and image support

4

Front-end & Backend development:

- next.js
- python

5

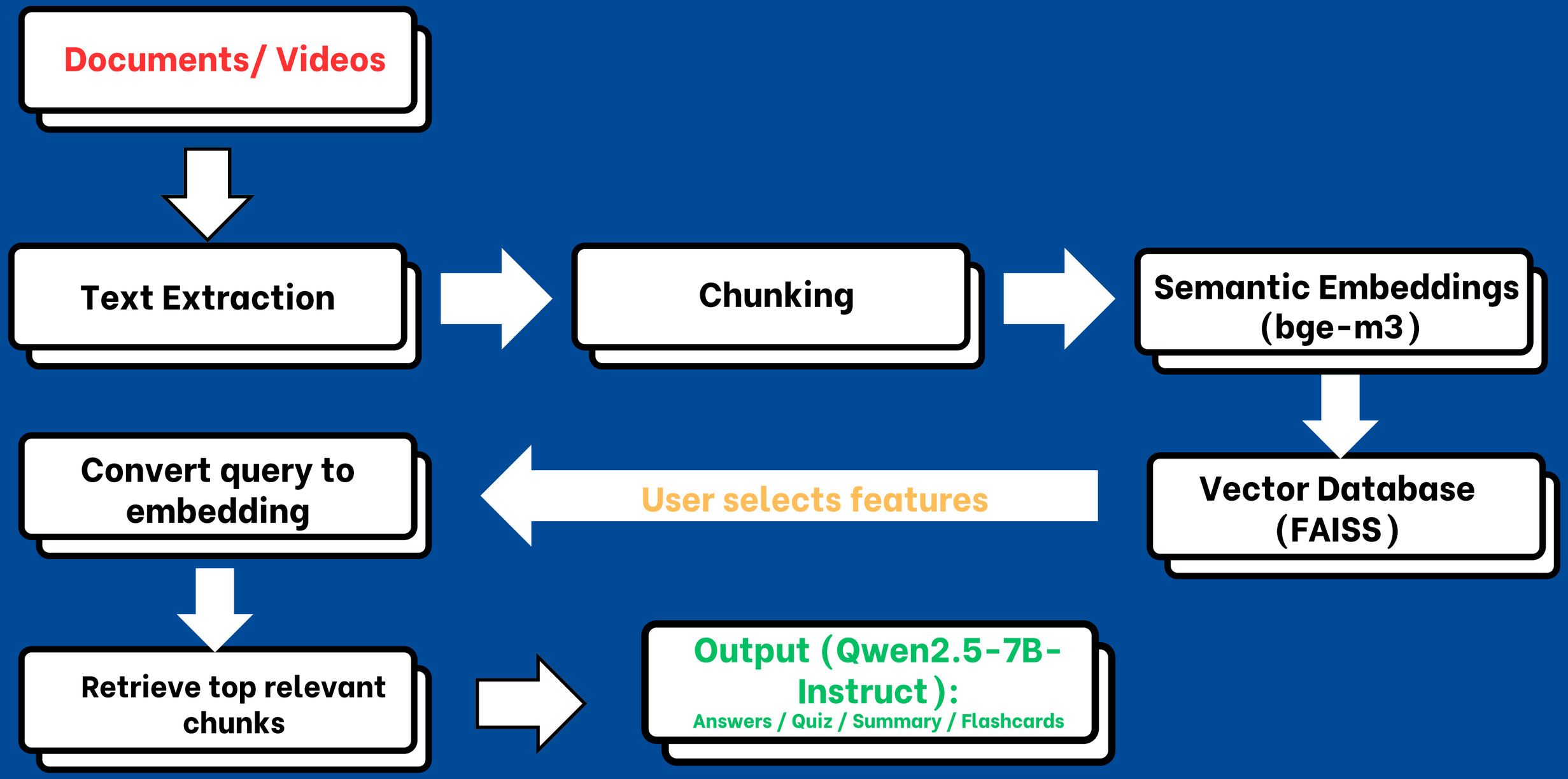
CLOUD AND CODE MANAGEMENT:

- AWS - EC2
- Github

RAG +Vector Database (FAISS)

- The model was losing *context* over time, so we introduced a **vector database** to store the context vectors.
- Store context as searchable embeddings
- Retrieve only relevant chunks at query time
- Feed focused context to the model (default top-k = 5, can be increased)

LearnLoop Pipeline



EXPLORATORY MODEL BENCHMARKING

Using Total time taken, tokens/sec, successful output generation, and output quality to evaluate performance

```
Question: what is the pdf about

--- BENCHMARK ---
Input tokens: 4637
Output tokens: 250
Tokenization time: 0.01s
Generation time: 11.88s
Total time: 11.90s
Tokens/sec: 21.05
The PDF document describes a comprehensive Natural Language Processing (NLP) course taught by Professor Amir Jafari at George Washington University. Key features include:

- **Course Description**: Provides foundational knowledge and advanced skills in NLP, transitioning from classical methods to deep learning and generative AI.

- **Learning Outcomes**:
  - Gain a foundational understanding of NLP methods and algorithms.
  - Master text mining and processing techniques.
  - Apply classical and deep learning models to practical AI applications.
  - Implement and fine-tune neural networks using PyTorch.
  - Design and deploy generative AI applications.
  - Evaluate and compare NLP and AI models across different problem domains.

- **Resources**: Includes textbooks, software setup tools, and cloud platforms (AWS, GCP).

- **Prerequisites**: Requires prior knowledge of Machine Learning I and Introduction to Data Mining.

- **Assessment**: Homework, quizzes, exams, and a final project.

- **Security Measures**: Encourages students to inform professors of absences and discusses policies regarding absences and tardiness.

- **Academic Integrity**: Emphasizes honesty and encourages students to avoid plagiarism and academic dishonesty
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Future Scope

- Personalized study plans
- Weak-topic detection
- Voice tutor mode
- Knowledge model (orchestrator/another model fine tuned on arxiv research papers)

LearnLoop Demo

Sources

How to Build PDF RAG Chatbot with LangChain, OpenAI, Streamlit & PGVector

https://huggingface.co/Qwen/Qwen2.5-7B-Instruct

https://huggingface.co/BAAI/bge-m3

chatgpt.com

Thank you!

-Raye, Kartik & Aditi